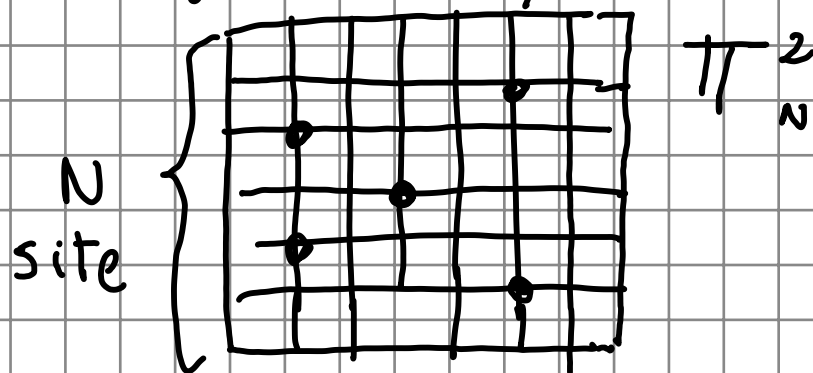


Introduction to interacting particle systems (IPS)

Toy model of non equilibrium on a lattice
(π_N^d)



$\eta(x)$ is the occupation variable at x ,
 $\eta(x) \in \{0, 1\}$

$\underline{\eta} = \{\eta(x)\}_x \in \Sigma_N$ configuration that evolves into

$\underline{\eta}(t) = \{\eta_t(x)\}_x$ with particles performing R.W.

along rules (hardcore) encoded in a Markov generator

These dynamics allows to derive laws (hydrodynamics) for macroscopic observables,

e.g.
$$\left| \pi_N(n, du) = \frac{1}{N!} \sum_x \eta(x) \delta_x(du) \right| \text{ empirical measure}$$

We can prove $\partial_t \rho = \partial_x (D(\rho) \partial_x \rho)$ as LLN.

Microscopic conservation law

$$\left| \eta_t(x) - \eta_0(x) + \overline{\nabla \cdot J_t(x)} = 0 \right| := \sum_{y \sim x} J_t(x, y)$$

$J_t(x, y)$ is the net flow of particles that crossed (x, y) in $[0, t]$.

Programme

-) Chap. 4 of KL
-) What was known + discrete exterior calculus
-) my results.